

Question Paper

Exam Date & Time: 02-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. MAKEUP DEGREE EXAMINATIONS - JANUARY 2026

Software Engineering [ECM_3124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) In a library management system case study, how would you conduct requirement analysis to capture both functional requirements (like book search, issue, and return) and non-functional requirements (such as security and performance)? (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) Define Software Requirement Specification (SRS) and explain its importance in software development. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) In the design of a library management system, how would you ensure good software design by applying principles of high cohesion and low coupling among the modules? What design approach would you choose and why? (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) What are the key characteristics of a good software design? (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) In the context of an online ticket booking system, describe the process of structured design after completing the DFD model. What modules and function hierarchies would be defined, and how would you ensure the design supports concurrency and real-time booking updates? (5)
- A)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) What is the purpose of design review in software design, and how does it improve the final product? (5)
- Course Outcome: "CO4" / Cognitive Level: "Remember"
- 4) Using UML, design interaction diagrams (sequence and collaboration diagrams) for the online ticket booking system, illustrating how users select seats, make payments, and receive bookings. (5)
- A)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) What are the basic concepts of object orientation in software modelling? (5)
- Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) For an online ticket booking system, discuss strategies for integration testing and system testing to ensure that modules like seat selection, payment processing, and ticket confirmation work correctly together. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Explain the difference between Black-Box Testing and White-Box Testing. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 ECM_3124:Software Engineering

#	Course Outcome	Description	Marks
1	2	CO 2	10
2	3	CO 3	10
3	4	CO 4	20
4	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 ECM_3124:Software Engineering

#	Level	No of Questions	Marks (%)
1	Remember	1	5(10%)
2	Understand	2	10(20%)
3	Application	3	15(30%)
4	Analysis	4	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)